

Two-Year Research Plan, August 2026 to July 2028

Estimating Social-Interaction Effects on Choice Behaviour Under Inferred Social Networks

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Item	Content
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How to read this document. The plan is set at two resolutions. Everything through December 2026 is planned at biweekly resolution, because that is the horizon that can currently be stood behind in detail. Everything from January 2027 onward is planned at quarterly resolution only: a biweekly plan for late 2027 or 2028 would be false precision rather than useful detail. The quarterly spine still names every deliverable, decision point, and conference target currently in view; it is refined one quarter at a time as each becomes current, and the biweekly section is extended in the same way as its own horizon is reached.

Dates that depend on something outside the author’s control, above all conference deadlines, are marked **to verify**. None of them have been confirmed against the relevant conference websites; they are carried forward at the precision actually known, rather than invented more precisely than they are known.

1 Research overview

1.1 Background

Discrete choice models with social-interaction effects extend standard choice models with a term for how a person’s choice is shaped by the choices or characteristics of the people around them, following Brock and Durlauf (2001) and, for the case where the relevant social structure

is a network rather than a single reference group, Lee, Li and Lin (2014). Estimating these models has, until recently, required the social network itself to be directly observed, in practice through a name-generator survey in which every person in the study population is asked to name the people they are tied to. Such surveys are expensive to run at any scale beyond a small sample, and for most existing mobility and travel-behaviour datasets they simply do not exist. Mobility data, by contrast, of the kind already collected for transportation research purposes, is comparatively abundant.

1.2 Problem statement

If a social network can instead be inferred computationally from mobility data, a choice model with social-interaction effects becomes usable on populations for which no elicited network data was ever collected. But every inference method makes errors, and those errors are not random noise: a network built from spatial co-location, for instance, will tend to systematically overstate ties between people who are frequently near each other but not socially connected, so that a network intended to represent social relationships risks being, to a real extent, a proximity network wearing a social-network label. The existing network-based choice estimator was built assuming the network is known exactly; applying it to a predicted network without accounting for that prediction error risks reporting a confident result that is actually an artefact of the inference procedure rather than a genuine social effect.

1.3 Objective and contribution

This research develops a method for **estimating social-interaction effects on choice behaviour under inferred social networks**: how to produce a usable network from mobility data, how to estimate a social-interaction choice model on that network, and how to assess how much the resulting estimates can be trusted given that the network was predicted rather than observed.

A design principle follows directly from this objective: network-inference methods that require no elicited social-tie data at all, from any population, are preferred over methods that need such data to calibrate. Methods that do require an elicited calibration sample are treated as one candidate family among several and used mainly as a benchmark for comparison, not as the default path. The aim is to push this as far as it will go: a framework whose core case needs no social-network survey of any kind, not merely no survey of the specific population being studied.

1.4 Research questions

RQ1 (primary). How should social-interaction effects in discrete-choice models be estimated, and their reliability assessed, when the social network is inferred from mobility data rather

than directly observed?

- **RQ1a.** What are the candidate methods for inferring a social network from mobility data, and what is each method’s error structure?
- **RQ1b.** How does network-inference error propagate into the endogenous and contextual social-interaction estimates of a network-based discrete choice model, and can that propagation be corrected or bounded?
- **RQ1c.** Can a genuine social effect be distinguished from a proximity artefact of the network-generation process, and what auxiliary information is needed to make that distinction?

RQ2 (applied case). Does a binary-choice version of this framework work end to end, and does it extend to a multinomial choice across community groupings without losing the properties, such as multiple equilibria and tipping behaviour, established at the binary stage?

- **RQ2a (Stage 1, binary).** For a binary choice, how do estimated endogenous and contextual social-interaction effects compare across networks generated from mobility data by different methods?
- **RQ2b (Stage 2, multinomial).** Does the estimation approach validated at Stage 1 extend to a multinomial choice across community groupings, and does the multiple-equilibria and tipping behaviour established in the binary case persist, weaken, or disappear as the number of alternatives grows?

2 Approach

2.1 Staged model development

The model is developed in two deliberate stages rather than attempted directly at the harder, multinomial level.

Stage 1, binary choice. A single yes-or-no choice is run through the existing social-interaction estimator with a network-generation method substituted in place of an observed network. This is the fastest path to a working, checkable pipeline, because the theory and the estimator already exist and do not need to be re-derived; only the network input and its consequences are new.

Stage 2, multinomial choice. Once Stage 1 runs cleanly end to end, the model is extended to a multinomial choice across community groupings, using whichever multinomial estimator for the network-interaction case the literature search turns up, or, if none exists, developing one as part of this research’s own contribution.

Staging the work this way separates two questions that would otherwise be tangled together: a problem discovered at Stage 2 is then legible as a multinomial-extension problem, not muddled together with a network-inference problem discovered at the same time.

2.2 Network-generation methods

Three families of method are under consideration, evaluated against the criteria below rather than one assumed by default:

- **Rule-based co-location or co-movement methods.** A fixed distance or time threshold applied to when two people are recorded near each other or moving together. Requires no elicited data of any kind. Its error is a known function of the threshold, which makes it the easiest family to reason about analytically.
- **Unsupervised network-formation or clustering methods.** Structure is inferred from the mobility data's own statistical patterns, without reference to any directly observed ties. Requires no elicited data.
- **Supervised tie-strength models.** A model trained on a sample of directly elicited tie strengths, then applied to unlabelled mobility data at scale. Requires an elicited calibration sample, though not of the population ultimately studied. Included as a benchmark against which the no-elicited-data methods above can be compared, not as the primary path.

Selection criteria, in order of priority for this research:

1. Does the method's error structure admit a tractable propagation into the choice-model estimates? An opaque generator is a liability here, since this propagation is the research's actual contribution.
2. Does it need elicited calibration data, and if so, how much and from whom?
3. Does it produce continuous weights, usable directly in the estimator, or only a hard tie or no-tie assignment, which would need an additional, separately justified binarizing decision?

2.3 Estimation methods

Four estimation approaches are relevant to this programme. All four solve the same underlying problem, that each person's belief about their neighbours' behaviour has to be consistent with what those neighbours actually do, but differ in how they arrive at that consistency.

- **NFXP (nested fixed point).** For every candidate parameter set, solve the network's belief system exactly, then compare to the data. Exact but comparatively slow. This is the approach already implemented and running for Stage 1.

- **NPL (nested pseudo-likelihood)**. Alternates between updating beliefs given the current parameters and updating parameters given the current beliefs, rather than fully re-solving the belief system at every step. Faster than NFXP, at some cost to the convergence guarantee.
- **MPEC (mathematical programming with equilibrium constraints)**. Hands parameters and beliefs to a single constrained optimizer together, with consistency of beliefs written in as a constraint rather than solved for separately. Can scale better for large problems, at the cost of needing a heavier-duty solver.
- **BLP (Berry, Levinsohn and Pakes)**. Built for settings where only aggregate shares are observed rather than individual choices. Relevant as a candidate for the multinomial Stage 2 case, where the object of interest may be closer to a group share than to one person’s choice.

Stage 1 does not require choosing among these; NFXP is already in use. The choice among NPL, MPEC, and BLP is a Stage 2 decision, to be made once the Stage 2 literature search is complete.

2.4 Data and validation practice

The primary data source is GPS mobility data. Access terms are still being finalized. Where real data is not yet available, model development proceeds on a synthetic or best-available fallback dataset, with every result that depends on real data marked accordingly until access is confirmed.

Every model component is derived by hand before it is coded, and every implementation is validated against its own hand derivation, and where an existing implementation is available, cross-checked against that implementation, before being applied to real or fallback data.

3 Programme structure

Paper 1 delivers the network-inference-and-estimation framework together with its applied case: a destination or activity choice across community groupings, reached through the two-stage path above.

Paper 2 (indicative, method not yet fixed) takes up the comparative question that Paper 1 sets aside: how much the social-interaction estimates depend on which network-generation method produced the network, tested by applying two or more different generation methods to the same population and comparing the resulting estimates. This question is deliberately not attempted within Paper 1, so that Paper 1’s contribution stands on a single, well-characterized method before the comparative question is opened.

4 Timeline

The interactive timeline is only available in the HTML version of this document, since it depends on a diagram renderer that is not guaranteed to render in this PDF build. The same information is in the quarterly and biweekly tables below.

5 Quarterly goals

Table 1: Quarterly goals, August 2026 to July 2028

Quar- ter	Months	Headline goal
Q1	Aug to Oct 2026	• Research plan finalized and discussed with the supervisor • Candidate network-generation methods identified and evaluated against the selection criteria below • Binary-choice (Stage 1) estimation engine ported to Python • First contact made with GPS mobility data, real or best available fallback
Q2	Nov 2026 to Jan 2027	• Stage 1 estimation running end to end on real or realistic fallback data • Stage 1 results written up • Literature search resolved on whether an existing multinomial network-interaction estimator can be reused
Q3	Feb to Apr 2027	• Stage 2 (multinomial) model formalized by hand • Estimation approach for Stage 2 selected (NPL, MPEC, or BLP) • Overseas internship shortlist drawn up and first contact made with candidate hosts
Q4	May to Jul 2027	• Stage 2 implemented and validated against its own hand derivation • Paper 1 drafting begins • Internship host and funding confirmed for a window between Oct 2027 and Mar 2028, or the internship set aside for this cycle

Quarter	Months	Headline goal
Q5	Aug to Oct 2027	• Paper 1 full draft completed • Internal review conducted • Conference presentation (hEART, Kraków, dates to verify; EASTS Adelaide kept as a fallback if the hEART abstract is not accepted) • Paper 2 scoping begins • If the Oct-Dec 2027 internship window was confirmed, the stay begins
Q6	Nov 2027 to Jan 2028	• Paper 1 submitted to a journal • Conference presentation (IATBR, Sydney, dates to verify) • Paper 2 methodology decided • Overseas internship underway, in whichever confirmed window (Oct-Dec 2027 or Jan-Mar 2028) applies
Q7	Feb to Apr 2028	• Paper 2 modelling underway • Dissertation chapters drafted from Paper 1, and from Paper 2 if the timeline allows • If the Jan-Mar 2028 internship window was confirmed, the stay concludes early in this quarter and dissertation drafting is adjusted accordingly
Q8	May to Jul 2028	• Dissertation compiled • Revisions from Paper 1 (and Paper 2, if under review) incorporated • Defense preparation

6 Biweekly detail, August to December 2026

Ten fortnights, Monday-anchored. This section is refreshed and extended a fortnight at a time as each one becomes current; it should be read as a plan for the next few months, not as a promise made about December from August.

Table 2: Biweekly detail, August to December 2026

#	Dates	Deliverables
1	Mon Aug 3 to Sun Aug 16	• Research plan and supporting documents finalized • Candidate network-generation methods surveyed • New Python repository scaffolded

#	Dates	Deliverables
2	Mon Aug 17 to Sun Aug 30	• Plan discussed with the supervisor • Python Stage 1 port validated against the existing implementation
3	Mon Aug 31 to Sun Sep 13	• First network-generation pass run on the best available data (real, if access is secured; a named fallback otherwise)
4	Mon Sep 14 to Sun Sep 27	• A network produced that the Stage 1 estimator can run on • Stage 1 results written up • Literature search on multinomial network-interaction estimators underway
5	Mon Sep 28 to Sun Oct 11	• Progress presented at a laboratory seminar
6	Mon Oct 12 to Sun Oct 25	• Post-presentation revisions incorporated
7	Mon Oct 26 to Sun Nov 8	• Stage 2 estimation approach chosen (NPL, MPEC, or BLP) • Stage 2 formalized by hand
8	Mon Nov 9 to Sun Nov 22	• Stage 2 code-porting begins from the hand derivation
9	Mon Nov 23 to Sun Dec 6	• Stage 2 code validated against the hand derivation on synthetic data
10	Mon Dec 7 to Sun Dec 20	• Stage 2 applied to real or best-available data • Interim results documented

7 Decision gates

Named, dated points where this plan genuinely branches, so a missed gate is visible rather than quietly absorbed into the next quarter.

Table 3: Decision gates

Gate	Target	If missed
Network-generation method selected for Stage 1	End of fortnight 2 (about Aug 30)	• Proceed with the simplest rule-based generator for Stage 1; revisit once a stronger candidate is confirmed usable

Gate	Target	If missed
GPS data access confirmed	End of fortnight 3 (about Sep 13)	Escalate directly to the supervisor; continued silence past this point is a signal to ask for help, not to keep waiting
Stage 1 running end to end on real (or realistic fallback) data	End of fortnight 3 (about Sep 13)	The October presentation shows the fallback-data result honestly, not a real-data result standing in for one that did not happen
Multinomial network-interaction estimator found in the literature, or claimed as this paper's originality	End of fortnight 4 (about Sep 27)	Feeds directly into how Stage 2 is framed from that point forward
Paper 1 full draft exists	End of Q4 (about Jul 2027)	Push back the Q5 submission target rather than submit before internal review is complete
Overseas internship host and funding confirmed	End of Q4 (about Jul 2027)	Set the internship aside for this dissertation cycle; domestic laboratory visits become the primary outside-feedback route instead

8 Dissemination plan

Venue	When	Submission deadline	Role
hEART, Kraków, Poland	2027 edition (exact dates to verify)	To verify	Primary target for Paper 1. A PhD school runs the day before the main symposium and is also worth attending
IATBR (International Association for Travel Behaviour Research), Sydney	December 2027 (to verify)	To verify	Paper 1, full version, once the journal submission is underway

Venue	When	Submission deadline	Role
EASTS (Eastern Asia Society for Transportation Studies), Adelaide	September 2027 (to verify)	To verify	Fallback only, if the hEART abstract is not accepted; not a parallel target

The 2027 Kraków dates above are projected from the confirmed 2026 edition, which was held in Paris (Université Gustave Eiffel, hosted at the Sorbonne conference cluster), 29 September to 1 October 2026, with an abstract deadline of 15 April 2026 and a PhD school on 28 September. The 2026 edition has already taken place; only its pattern, not its dates, carries forward to 2027.

8.1 Research visits and overseas internship

Suggestions from the supervisor are genuinely welcome on either list below, the domestic visit targets or the overseas internship host: neither is meant to be closed.

8.1.1 Domestic laboratory visits

Independently of the conference targets above, shorter, lower-stakes visits to other laboratories for feedback are worth keeping open as options, presented in English:

- The University of Tokyo, Giancarlos Sensei’s laboratory
- Kochi University of Technology, Urban Network Design Laboratory
- Other Japanese universities, to be identified

8.1.2 Overseas research internship

A research internship of two to three months at a university outside Japan is a further objective, following the precedent of a labmate’s own stay abroad earlier in 2026. The target window is left open between **October to December 2027** and **January to March 2028**; the later option overlaps the dissertation-drafting quarter and would require that quarter’s work to be pulled forward if chosen. Typical lead time for host agreement and funding is six to twelve months, so the host search begins well ahead of either window.

Three kinds of host group are being considered, at the level of institution rather than individual, since no host has yet been contacted:

- **Network inference and computational social science**, at institutions with strong programmes inferring social structure from mobility or proximity data, such as the Technical University of Denmark and Northeastern University’s Network Science Institute.
- **Econometrics of social interactions and network formation**, at institutions working on models where the network itself is estimated rather than assumed known, such as National Taiwan University and Monash University.
- **Transport and choice modelling**, at institutions with an established travel-behaviour and discrete-choice research group, such as the National University of Singapore, the University of Washington, and the Choice Modelling Centre at the University of Leeds.

The selection criterion is that the host should strengthen a part of the framework that is harder to strengthen from Hiroshima alone.

For reference, the candidates currently identified in each cluster, none yet approached:

- **Network inference and computational social science.** Sune Lehmann (Technical University of Denmark, Copenhagen Networks Study) and Esteban Moro (Northeastern University, Network Science Institute). DTU alone may cover most of the framework’s threads, since Laura Alessandretti (human mobility) and Francisco Pereira (transport machine learning) are also there.
- **Econometrics of social interactions and network formation.** Chih-Sheng Hsieh (National Taiwan University; Lung-fei Lee’s own co-author, already cited in the reading list) and Yves Zenou (Monash University; network economics and peer effects).
- **Transport and choice modelling.** Prateek Bansal (National University of Singapore, BeCoS lab; the labmate’s own precedent), Cynthia Chen (University of Washington, THINK lab), and the Choice Modelling Centre at the University of Leeds.

Affiliations were verified by search in August 2026 and have not been re-confirmed since; whether each group currently accepts visiting doctoral students has not been checked and is the first thing to find out before approaching anyone.

9 Risks and dependencies

1. **An inferred network can encode geographic proximity rather than genuine social relationship.** Whichever generation method is used, this has to be checked for explicitly, not assumed away, since a network built substantially from distance features will produce estimates that look like a social effect but are not.
2. **The network is predicted, not known, and the underlying estimator assumes it is known.** This is the research’s actual contribution rather than a caveat, but it means every result carries an extra layer of uncertainty that a standard application of the estimator would not have to report.

3. The GPS dataset has not been inspected first-hand as of this writing.

- Ownership of the dataset and its licence terms are not yet fully confirmed.
- Everything in this plan that depends on real data is provisional on access being granted on a timeline compatible with the plan above.
- See the decision gate at fortnight 3 for the fallback if access is delayed.

10 Plan maintenance

This plan is refreshed at the start of each quarter, and the biweekly section is extended a fortnight at a time as its own horizon is reached, so the document stays honest about what is actually known rather than fixed at the level of detail available on the day it was written.